

Comparative Analysis of Time Series Forecasting Models for Cabai Rawit Merah Price Prediction in Central Jakarta

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Abstract—Bird's eye chili is a major horticultural commodity in Indonesia whose prices frequently fluctuate due to changes in supply, demand, and overall market conditions. Accurate forecasting is therefore important for supporting production planning, distribution management, and food price stabilization. This study examines the behavior of daily bird's eye chili prices in Central Jakarta and compares the forecasting performance of several time series methods. The analysis uses 1,305 daily observations obtained from the National Strategic Food Price Information Center (PIHPS) covering the period from May 2021 to May 2026. The evaluated models include Naïve Forecasting, Exponential Smoothing (ETS), Autoregressive Integrated Moving Average (ARIMA), Seasonal ARIMA (SARIMA), Time Series Regression, and Neural Network Autoregression (NNAR). Prior to model construction, the dataset was explored and preprocessed through Box-Cox transformation, differencing, and stationarity testing. Forecast accuracy was measured using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). The findings show that the series is characterized by considerable price volatility, temporal dependence, and relatively weak seasonal effects. Among the evaluated models, NNAR(2,0,2) delivered the best predictive performance, producing an RMSE of 22,311, an MAE of 17,039, and a MAPE of 21.86%, thereby outperforming ETS, ARIMA, SARIMA, and lag-based regression models. These results indicate that nonlinear forecasting techniques are better suited to representing the complex dynamics of bird's eye chili prices and can provide useful insights for anticipating future market fluctuations.

Keywords—bird's eye chili price, time series forecasting, ARIMA, SARIMA, neural network autoregression, price prediction.

I. INTRODUCTION

Bird's eye chili (*Capsicum frutescens*) is one of the most widely consumed horticultural commodities in Indonesia and serves as an essential ingredient in daily food consumption. Due to its high demand and economic importance, fluctuations in bird's eye chili prices often receive considerable attention from consumers, producers, and policymakers [1], [2]. Maintaining price stability is therefore important not only for market participants but also for broader economic and food security concerns.

Despite its importance, bird's eye chili prices are highly volatile and can change significantly within short periods. Price fluctuations are influenced by various factors, including seasonal production cycles, weather conditions, supply chain disruptions, and sudden increases in demand during religious

holidays and national events [3], [4]. Moreover, substantial increases in chili prices have been identified as one of the contributors to food inflation in Indonesia, potentially affecting household purchasing power and market stability [5]. Such volatility creates uncertainty in the market and highlights the need for reliable forecasting methods that can support decision-making and price stabilization efforts.

Central Jakarta is one of the main food consumption and distribution centers in Indonesia, where traditional markets play an important role in connecting producers and consumers. Changes in bird's eye chili prices in these markets can directly influence purchasing behavior, trading activities, and local market dynamics. Given the strategic role of Central Jakarta in the food supply chain, anticipating future price movements is valuable for improving market efficiency and supporting government initiatives aimed at maintaining food price stability.

Forecasting provides a practical approach for estimating future prices using historical observations. Accurate forecasts can support production planning, inventory management, distribution strategies, and policy interventions designed to mitigate the impact of price shocks [6]–[8]. However, forecasting agricultural commodity prices remains challenging because such data often exhibit trend, seasonality, irregular fluctuations, and nonlinear patterns. Understanding these underlying characteristics is an important preliminary step before model development, as the effectiveness of a forecasting approach depends heavily on the patterns present in the data.

Previous studies have investigated chili price forecasting using a wide range of approaches, including statistical models, fuzzy-based methods, machine learning algorithms, and deep learning architectures [9]–[13]. Although these studies demonstrate promising forecasting performance, most comparisons are conducted only within the same methodological family. Consequently, it remains unclear whether conventional statistical models, regression-based approaches, or nonlinear neural-network models provide superior forecasting performance when evaluated under identical experimental conditions using the same dataset.

As a result, limited evidence is available regarding the comparative performance of statistical, regression-based, and neural-network forecasting approaches when evaluated under a unified experimental framework using long-term daily bird's eye chili price data. Furthermore, studies focusing on traditional market prices in Central Jakarta remain relatively scarce despite the region's strategic role in Indonesia's food

distribution network. To address these gaps, this study presents a comprehensive comparison of six forecasting families, namely Naïve Forecasting, Exponential Smoothing (ETS), ARIMA, SARIMA, Time Series Regression, and Neural Network Autoregression (NNAR), using 1,305 daily price observations obtained from the National Strategic Food Price Information Center (PIHPS) between May 2021 and May 2026 [16]. By evaluating benchmark, statistical, regression-based, and nonlinear forecasting models under identical experimental conditions, this study provides a clearer understanding of their relative strengths for modeling highly volatile agricultural commodity prices. Forecasting performance is assessed using RMSE, MAE, and MAPE to identify the most suitable model for predicting bird's eye chili prices in Central Jakarta.

II. RELATED WORK

Forecasting agricultural commodity prices has attracted considerable attention due to its importance in supporting production planning, supply chain management, and policy decision-making. Various forecasting approaches have been applied to predict chili prices, ranging from traditional statistical methods to machine learning and deep learning techniques. However, forecasting performance often depends on dataset characteristics, forecasting horizons, and modeling assumptions, making it difficult to generalize the effectiveness of a particular method across different applications [5]–[14].

TABLE I. SUMMARY OF PREVIOUS STUDIES ON CHILI PRICE FORECASTING

Study	Data & Location	Method	Main Findings	Limitation
Setiawan et al. [13]	Monthly bird's eye chili prices, Mataram	ARIMA	ARIMA(2,0,2) produced the best forecasting performance	Evaluates only ARIMA models
Arisandi et al. [3]	Daily bird's eye chili prices, Makassar	Single Exponential Smoothing (SES)	Achieved MAPE of 6.50%	Uses a single statistical method
Saphira et al. [14]	Chili price data	Double and Triple Exponential Smoothing	TES outperformed DES	Limited to exponential smoothing methods
Komaria et al. [10]	Weekly bird's eye chili prices, East Java	Fuzzy Time Series Lee	Achieved MAPE of 13.09%	Focuses only on fuzzy-based forecasting
Anggara et al. [11]	Red bird's eye chili prices, West Java	LSTM and Attention-LSTM	Attention-LSTM achieved MAPE of 2.74%	Comparison limited to deep learning models
Anamsyah et al. [12]	Red bird's eye chili prices, East Java	XGBoost, LSTM, and Neural Prophet	LSTM achieved MAPE of 2.13%	Focuses only on machine learning and deep learning methods
Nasirudin and Dzikrullah [16]	Indonesian chili prices	Seasonal ARIMAX	Seasonal ARIMAX effectively captured seasonal price patterns	Limited to statistical modeling approaches
Manurung et al. [17]	Bird's eye chili prices	Hybrid ARIMAX-ANN	Hybrid model improved forecasting accuracy compared to individual models	Focuses on hybrid statistical-neural approaches
Darmawan et al. [18]	Chili price data	Bidirectional-LSTM and SSA	Bidirectional-LSTM demonstrated superior predictive performance	Comparison limited to selected advanced models

As shown in Table I, previous studies have demonstrated the effectiveness of various forecasting approaches for chili price prediction. Statistical methods such as ARIMA, Exponential Smoothing, and ARIMAX remain widely used due to their simplicity and interpretability [3], [13], [14], [16]. Meanwhile, recent studies have increasingly explored neural network and deep learning approaches, including LSTM, Bidirectional-LSTM, and hybrid models, which often provide higher predictive accuracy when nonlinear patterns are present in the data [11], [12], [17], [18].

Despite these advances, most existing studies compare forecasting models within the same methodological family. Statistical models are commonly evaluated against other statistical models, while machine learning and deep learning approaches are often assessed separately [11]–[14]. Consequently, limited evidence exists regarding the relative performance of forecasting approaches from different paradigms when applied under the same experimental setting. Furthermore, many previous studies employ weekly or monthly observations [10], [13], which may not fully capture short-term price fluctuations in agricultural commodity markets. Research utilizing long-term daily bird's eye chili price data from traditional markets in Central Jakarta also remains limited. Therefore, this study addresses these gaps by comparing benchmark, statistical, regression-based, and neural-network forecasting approaches using a unified dataset and evaluation framework. The objective is to provide a more comprehensive understanding of model performance for forecasting highly volatile bird's eye chili prices.

III. RESEARCH METHODOLOGY

A. Research Framework

Figure 1 illustrates the overall research framework employed in this study. The research process consists of six main stages: data collection, data preprocessing, exploratory data analysis, stationarity assessment, forecasting model development, and performance evaluation. The final stage identifies the forecasting model that provides the lowest prediction error and the most accurate forecasts for bird's eye chili prices in Central Jakarta.

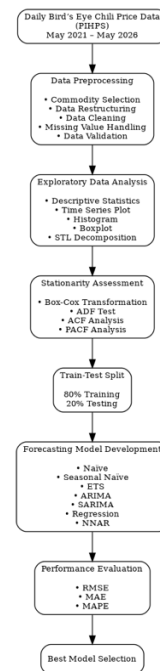


Fig. 1. Research Framework of the proposed forecasting model comparison for bird's eye chili price prediction in Central Jakarta

B. Dataset Description

The analysis was conducted using daily bird's eye chili price data retrieved from the National Strategic Food Price Information Center (PIHPS), an official price monitoring platform maintained by Bank Indonesia. The dataset represents traditional market prices in Central Jakarta and covers the period from May 2021 to May 2026, comprising a total of 1,305 observations.

Central Jakarta was chosen as the study area because it functions as one of Indonesia's primary food consumption and distribution hubs. As a result, changes in bird's eye chili prices can directly affect trading activities, consumer purchasing decisions, and the stability of local food markets. Furthermore, daily observations provide a finer representation of short-term price movements and volatility than weekly or monthly data.

C. Data Preprocessing

Before developing the forecasting models, several preprocessing steps were carried out to improve the quality and consistency of the dataset. These steps involved selecting the target commodity, restructuring the raw data, cleaning the dataset, handling missing values, and validating the processed data.

Following the preprocessing stage, the dataset was transformed into a univariate time series format appropriate for forecasting purposes. Because price records were available only on weekdays (Monday through Friday), the seasonal period was set to five observations. This configuration was then applied when developing the seasonal forecasting models.

For model evaluation, the dataset was partitioned into training and testing subsets using an 80:20 split. The training subset was utilized to build the forecasting models and estimate their parameters, whereas the testing subset was retained for out-of-sample performance assessment.

D. Exploratory Data Analysis and Stationarity Assessment

Exploratory Data Analysis (EDA) was performed to examine the underlying characteristics of the bird's eye chili price series. The analysis consisted of descriptive statistics, time series visualization, histogram and boxplot inspection, as well as seasonal-trend decomposition to better understand the behavior of the data.

The stationarity assessment began with a Box-Cox transformation to stabilize the variance of the series. Mean stationarity was then evaluated by applying differencing followed by the Augmented Dickey-Fuller (ADF) test. In addition, the autocorrelation structure was examined through Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots to identify temporal dependence and assist in selecting appropriate forecasting models.

The findings from these analyses provided a clearer understanding of the underlying behavior of the time series and served as the basis for developing the forecasting models evaluated in this study.

E. Forecasting Model Development

This study evaluated forecasting approaches from different methodological paradigms. Multiple candidate models were developed within each forecasting family, and

the best-performing model from each family was selected for final comparison.

1) Benchmark Models

To establish a baseline for comparison, this study employed both Naïve and Seasonal Naïve Forecasting models. The Naïve approach generates the forecast for the next period by using the latest observed value as the prediction.

$$\widehat{y}_{t+1} = y_t \quad (1)$$

In this formulation, y_t denotes the observed value at time t , whereas $\widehat{y}_{t+1}$ represents the forecast for the subsequent period.

Unlike the standard Naïve model, the Seasonal Naïve approach produces forecasts by repeating observations from the corresponding period in the previous seasonal cycle.

2) Exponential Smoothing Models

Exponential Smoothing (ETS) forecasts future values by assigning greater importance to more recent observations while progressively reducing the influence of older data. Owing to its ability to capture level, trend, and seasonal patterns, ETS has been widely adopted in agricultural commodity forecasting [8], [14].

Multiple ETS specifications with different smoothing parameters were examined, together with an automatically optimized configuration. The comparison was intended to identify the smoothing structure that provides the most accurate forecasts for bird's eye chili prices.

3) Arima Models

Autoregressive Integrated Moving Average (ARIMA) is a classical statistical approach that has been extensively applied to forecasting agricultural commodity prices. The model integrates autoregressive (AR), differencing (I), and moving average (MA) components into a single forecasting framework.

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j} \quad (2)$$

where (p) , (d) , and (q) denote the autoregressive, differencing, and moving average orders, respectively; ϕ_i and θ_j are model parameters; and ε_t represents the random error term.

Before fitting the ARIMA model, a Box-Cox transformation was performed to stabilize the variance of the series and improve the robustness of the forecasting model.

4) SARIMA Models

Seasonal Autoregressive Integrated Moving Average (SARIMA) extends the conventional ARIMA framework by explicitly incorporating seasonal patterns into the forecasting model.

$$\text{SARIMA}(p, d, q)(P, D, Q)_s$$

Where (p, d, q) represent the non-seasonal orders, (P, D, Q) represent the seasonal orders, and s denotes the seasonal period.

Since the dataset includes observations only on weekdays, the seasonal length was specified as $s = 5$, representing one complete weekly cycle.

5) Time Series Regression Models

Time Series Regression predicts future values by using previous observations of the target variable as explanatory variables.

$$y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \dots + \beta_n y_{t-n} + \varepsilon_t \quad (3)$$

Where β_i represent the regression coefficients and ε_t is the random error term.

Several regression models with different lag structures were evaluated through recursive forecasting, in which each predicted value was subsequently used to generate forecasts for the following time step.

6) Neural Network Autoregression Models

Neural Network Autoregression (NNAR) is a feed-forward neural network designed for time series forecasting by using lagged observations as input features. In contrast to conventional statistical models, NNAR is capable of learning complex nonlinear relationships directly from historical observations.

To determine the most suitable neural network structure, several NNAR architectures with different numbers of lagged inputs and hidden neurons were evaluated. An automatically optimized NNAR configuration was also considered, and the final model was selected according to its forecasting performance.

F. Performance Evaluation

The predictive performance of each forecasting model was assessed using three commonly adopted evaluation metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE)

RMSE is calculated as

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \quad (4)$$

MAE is defined as

$$MAE = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \quad (5)$$

And MAPE is calculated as

$$MAPE = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right| \quad (6)$$

Where y_t represent the actual value, $\hat{y}_t$ denotes the predicted value, and n is the total number of observations.

The forecasting model that produced the smallest RMSE, MAE, and MAPE values on the testing dataset was selected as the most appropriate approach for predicting bird's eye chili prices in Central Jakarta.

IV. RESULTS AND DISCUSSION

This section summarizes the findings obtained from the exploratory analysis, stationarity assessment, forecasting model development, and model evaluation. The discussion follows the research objectives by examining the characteristics of bird's eye chili price movements, identifying the most suitable forecasting model, and comparing the predictive accuracy of the evaluated approaches.

A. Characteristics of Bird's Eye Chili Price Series

Examining the characteristics of the price series is an essential preliminary step in forecasting because the patterns observed in historical data influence both the selection and predictive performance of forecasting models. The analysis was conducted using 1,305 daily observations of bird's eye chili prices obtained from traditional markets in Central Jakarta over the period from May 2021 to May 2026. A summary of the descriptive statistics is presented in Table I.

TABLE II. DESCRIPTIVE STATISTICS OF BIRD'S EYE CHILI PRICES

Statistic	Value
Observations	1305
Minimum	30,850
1st Quartile	50,000
Median	58,950
Mean	65,952
3rd Quartile	79,250
Maximum	136,250

The descriptive measures reveal considerable variation in bird's eye chili prices throughout the study period. The average price was IDR 65,952 per kilogram, with values ranging from IDR 30,850 to IDR 136,250. This wide range reflects the highly volatile nature of the commodity, which is commonly influenced by production cycles, weather conditions, transportation constraints, and fluctuations in consumer demand [1], [2].

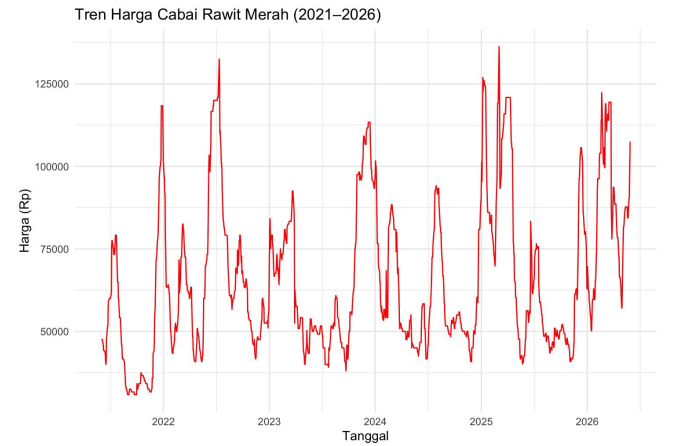


Fig. 2. Presents the daily movement of bird's eye chili prices throughout the observation period.

Fig. 2 illustrates the movement of bird's eye chili prices throughout the study period. The price series demonstrates considerable volatility, with repeated episodes of sharp increases followed by noticeable declines. Several pronounced price spikes can be observed, suggesting temporary market imbalances that were likely associated with supply shortages or unexpected increases in demand. Despite the high volatility, the series exhibits clear temporal dependence, indicating that current price movements remain influenced by historical observations. These characteristics justify the application of time series forecasting techniques, which rely on historical observations to estimate future price behavior.

To further investigate the underlying structure of the series, seasonal-trend decomposition was performed.

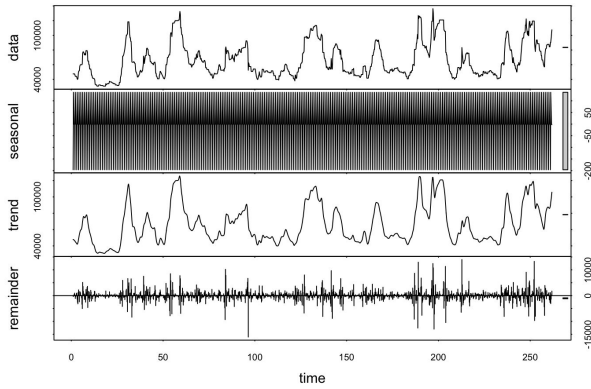


Fig. 3. STL decomposition of the bird's eye chili price series.

The STL decomposition separates the series into three primary components: trend, seasonal, and residual variations. The extracted trend component shows that price movements are systematic rather than completely random, displaying periods of sustained increases and declines throughout the observation period. The seasonal component shows recurring fluctuations that may be associated with weekly trading patterns and recurring market conditions. However, the seasonal amplitude appears relatively small compared with the overall variation of the series. The residual component exhibits considerable irregular variation, implying that a substantial portion of price fluctuations may be driven by external factors that are not explicitly represented in the forecasting models.

Overall, the exploratory analysis indicates that bird's eye chili prices in Central Jakarta are characterized by high volatility, temporal dependence, and moderate seasonal behavior. These characteristics justify the evaluation of both statistical and nonlinear forecasting models in the subsequent analysis.

B. Stationarity Assessment

Stationarity is a fundamental assumption for many statistical forecasting methods, particularly ARIMA-based models. Therefore, stationarity assessment was conducted to ensure that the underlying statistical properties of the series remained relatively stable over time.

Variance stationarity was first examined using the Box-Cox transformation. The estimated Box-Cox parameter was $\lambda = -0.3426$, indicating that the original series exhibited non-constant variance. Consequently, a Box-Cox transformation was applied to stabilize the variance and reduce the influence of extreme price fluctuations.

Following variance stabilization, stationarity in the mean was evaluated through differencing and formal statistical testing. The Augmented Dickey-Fuller (ADF) test produced a test statistic of -5.266 with a p-value below the significance threshold, indicating that the null hypothesis of a unit root could be rejected. Therefore, the transformed series can be considered stationary.

To support the stationarity assessment and model identification process, autocorrelation analysis was conducted using the ACF and PACF functions.

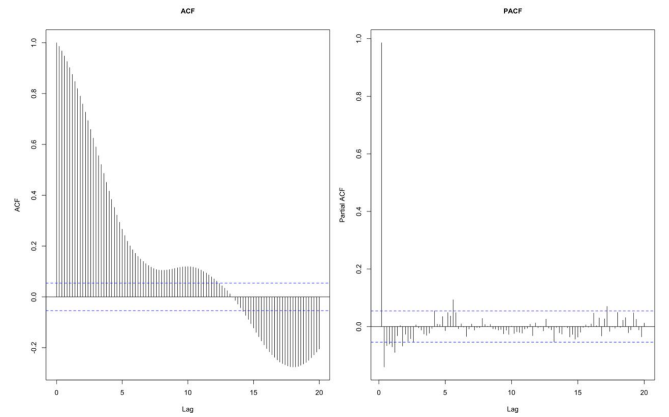


Fig. 4. ACF and PACF plots of the transformed series.

The ACF plot displays a gradual decay in autocorrelation, whereas the PACF plot is characterized by a prominent spike at the first lag followed by relatively minor partial autocorrelations. This autocorrelation pattern indicates the existence of short-term autoregressive dependence and provides initial evidence supporting the application of ARIMA-based forecasting models. The observed autocorrelation structure also indicates that historical observations contain meaningful information that can be exploited for forecasting purposes.

Based on the variance stabilization, differencing procedure, and ADF test results, the transformed series satisfies the stationarity requirements necessary for subsequent forecasting analysis.

C. Forecasting Model Development

To identify the most suitable forecasting technique, this study compared several forecasting families, including benchmark, exponential smoothing, autoregressive, regression-based, and neural network models. Multiple candidate models were developed within each category, and their performance was compared using accuracy metrics calculated from the testing dataset.

The best-performing model from each forecasting family is presented in Table III.

TABLE III. BEST MODEL FROM EACH FORECASTING FAMILY

Forecasting Family	Selected Model
Naïve	Naïve
Exponential Smoothing	ETS (0.5,0.5,0.4)
ARIMA	ARIMA (1,0,0)
SARIMA	SARIMA (1,0,0) (1,0,0) [5]
Regression	Regression (lag1)
Neural Network	NNAR (2,0,2)

Within the benchmark category, the Naïve model produced the strongest performance. Despite its simplicity, it serves as an effective baseline because each prediction is generated solely from the latest observed value. However, its relatively large forecasting errors suggest that the dynamics of bird's eye chili prices cannot be adequately explained using only the persistence assumption.

Among the exponential smoothing models, ETS (0.5,0.5,0.4) produced the best predictive performance. This result indicates that smoothing the level, trend, and seasonal components enables the model to generate more accurate forecasts than the benchmark method.

Within the autoregressive model family, ARIMA (1,0,0) was identified as the best-performing specification. The selection of a first-order autoregressive model suggests that recent observations have a strong influence on subsequent price movements. More complex ARIMA configurations did not provide additional forecasting improvements, indicating that a simpler specification was sufficient to capture the temporal dependence present in the data.

Among the seasonal forecasting models, SARIMA (1,0,0)(1,0,0)[5] achieved the highest accuracy. The seasonal period of five corresponds to the weekday trading cycle represented in the dataset. Nevertheless, incorporating seasonal components resulted in only a marginal improvement in predictive performance compared with the non-seasonal model.

Within the regression family, the lag-1 regression model achieved the highest accuracy. This result further confirms the importance of short-term temporal dependence in bird's eye chili prices.

Finally, NNAR (2,0,2) was selected as the best neural network architecture. By combining lagged observations with nonlinear learning capabilities, the model can capture relationships that may not be adequately represented by conventional linear approaches.

D. Comparative Forecasting Performance

The selected forecasting models were compared using RMSE, MAE, and MAPE as evaluation metrics. A summary of the forecasting results is presented in Table IV.

TABLE IV. FORECASTING PERFORMANCE COMPARISON

Model	RMSE	MAE	MAPE (%)
Naïve	35,330	27,388	33.18
ETS (0.5,0.5,0.4)	24,542	18,730	24.33
ARIMA (1,0,0)	25,780	19,125	23.29
SARIMA (1,0,0) (1,0,0) [5]	25,375	19,049	23.56
Regression (lag1)	23,244	17,948	23.20
NNAR (2,0,2)	22,311	17,039	21.86

The results reveal substantial differences in forecasting performance among the evaluated models. The Naïve model produced the largest forecasting errors, with a MAPE of 33.18%. This result indicates that future bird's eye chili prices cannot be accurately predicted using only the most recent observation and that additional temporal structures must be considered.

The ETS model significantly improved forecasting accuracy compared with the benchmark model, reducing the MAPE to 24.33%. This improvement demonstrates the usefulness of smoothing techniques for capturing underlying patterns in the data. Nevertheless, ETS remained less accurate than ARIMA, regression, and neural network approaches.

ARIMA (1,0,0) achieved a MAPE of 23.29%, while SARIMA (1,0,0) (1,0,0) [5] obtained a slightly higher MAPE of 23.56%. The fact that ARIMA outperformed SARIMA suggests that seasonal effects contribute relatively little to predictive performance. Although seasonal fluctuations were observed during decomposition analysis, these patterns were not sufficiently strong to generate substantial forecasting improvements. This finding indicates that short-term autoregressive relationships play a more important role than seasonal behavior in explaining bird's eye chili price movements.

The lag-based regression model achieved a MAPE of 23.20%, slightly outperforming both ARIMA and SARIMA. This result demonstrates that a simple linear relationship between current and previous observations can explain a considerable proportion of the variability in bird's eye chili prices.

The best overall performance was achieved by NNAR (2,0,2), which produced the lowest RMSE, MAE, and MAPE values among all evaluated models. The superior performance of NNAR suggests the presence of nonlinear dependencies within the series that cannot be fully captured by traditional linear forecasting approaches.

E. Evaluation of the Best Forecasting Model

Since NNAR (2,0,2) achieved the highest forecasting accuracy among all competing models, a more detailed evaluation was conducted to assess its predictive capability.

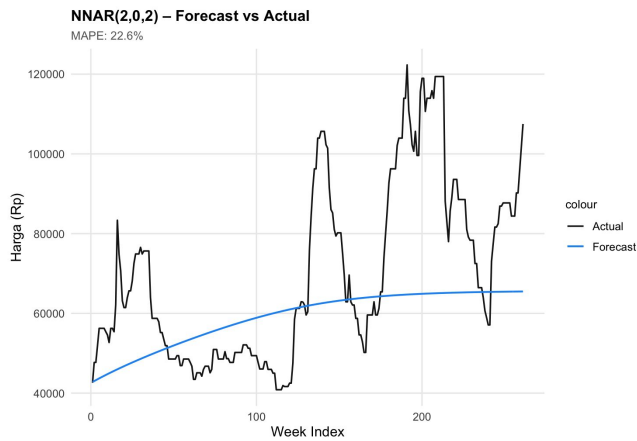


Fig. 5. Actual and forecasted bird's eye chili prices generated by NNAR (2,0,2).

Fig. 5 compares the actual observations and forecasts generated by NNAR (2,0,2) on the testing dataset. The figure shows that the model successfully follows the general direction of price movements and captures many of the short-term fluctuations present in the series. Forecasted values closely track actual observations during periods of relatively stable prices, indicating that the model effectively learns the temporal dependencies embedded in the data.

However, some deviations can be observed during periods characterized by abrupt price spikes and sudden declines. Such events are difficult to predict because they are often influenced by external factors that are not explicitly included in the forecasting model, such as extreme weather conditions, supply chain disruptions, or unexpected demand shocks. Despite these challenges, NNAR (2,0,2) consistently produced lower forecasting errors than all competing models.

The superior performance of NNAR is consistent with previous studies demonstrating the effectiveness of neural-network-based forecasting approaches for agricultural commodity prices [5], [6], [11], [12], [17], [18]. Unlike linear models, neural networks can learn nonlinear relationships and complex interactions among historical observations. Consequently, NNAR was better able to capture the dynamic behavior of bird's eye chili prices in Central Jakarta.

Overall, the forecasting results indicate that NNAR (2,0,2) is the most suitable model among the evaluated approaches for predicting daily bird's eye chili prices. The model achieved an RMSE of 22,311, an MAE of 17,039, and a MAPE of 21.86%, demonstrating its ability to provide relatively accurate forecasts despite the highly volatile nature of the commodity price series.

V. CONCLUSION

This study analyzed daily bird's eye chili price data from traditional markets in Central Jakarta covering the period from May 2021 to May 2026. The exploratory analysis revealed that the price series exhibits substantial volatility, characterized by frequent fluctuations, occasional price spikes, and strong temporal dependence between consecutive observations. Decomposition analysis indicated the presence of trend and seasonal components; however, the seasonal

effect was relatively weak compared with the overall variability of the series. The stationarity assessment further confirmed that variance stabilization and differencing were required prior to the application of statistical forecasting models.

To identify the most suitable forecasting approach, six forecasting families were evaluated, namely Naïve, Exponential Smoothing (ETS), ARIMA, SARIMA, Time Series Regression, and Neural Network Autoregression (NNAR). The results showed that forecasting performance varied considerably across model families. While ETS, ARIMA, SARIMA, and lag-based regression models provided competitive results, the Naïve model produced the largest forecasting errors. The comparison also revealed that the inclusion of seasonal components did not significantly improve forecasting accuracy, as ARIMA slightly outperformed SARIMA. This finding suggests that short-term autoregressive behavior contributes more strongly to bird's eye chili price movements than seasonal effects.

Among all evaluated models, NNAR (2,0,2) achieved the best forecasting performance with an RMSE of 22,311, an MAE of 17,039, and a MAPE of 21.86%. The superior performance of NNAR indicates that the bird's eye chili price series contains nonlinear relationships that can be more effectively captured by neural-network-based approaches than by conventional linear models. Therefore, NNAR (2,0,2) can be considered the most suitable forecasting model for predicting daily bird's eye chili prices in Central Jakarta within the scope of this study.

Although the proposed approach demonstrated satisfactory forecasting performance, this study is limited to univariate price observations and does not incorporate external factors that may influence market dynamics. Future research may explore the integration of exogenous variables such as weather conditions, rainfall, production levels, transportation costs, or holiday effects. In addition, more advanced machine learning and deep learning approaches, including LSTM, GRU, Transformer-based models, or hybrid forecasting frameworks, may be investigated to further improve forecasting accuracy and support more effective decision-making in agricultural commodity markets.

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